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CrossFusionSleepNet: A multimodal deep learning model for automatic sleep stage classification

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A R T I C L E I N F O

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A B S T R A C T

Sleep staging is used to assess sleep quality and diagnose sleep-related disorders by monitoring and analyzing physiological signals during sleep. Although previous studies have demonstrated notable success in sleep staging, most current methods rely predominantly on single-modality or single-channel data. To achieve high accuracy in automatic sleep staging, this study proposes a multimodal approach that leverages different modality features in the time and frequency domains of multichannel signals. A parallel network combined with cross-attention mechanism was designed to facilitate the effective integration of multimodal features. The cross-attention mechanism captures the complex relationships between the time domain and frequency domain, thereby improving the performance of the parallel network. The CrossFusionSleepNet achieved accuracies of 89.0%, 87.5%, and 88.9%, with corresponding F1-scores of 84.0%, 83.0%, and 81.8% on multiple public sleep stage datasets (SleepEDF-20, SleepEDF-78, and SHHS). Our results demonstrate that the multimodal approach shows potential advantages over single-modality methods, offering a novel perspective for accurate automatic sleep staging.

1. Introduction

Sleep is an essential physiological process in humans. As sleep quality is directly linked to the immune system and emotional regulation [1], sleep disorders can result in low-quality sleep that interferes with daily activities [2], and may increase the risk of cardiovascular disease and mortality [3]. Sleep staging offers critical clinical value, as it can be used to monitor sleep quality and diagnose sleep-related disorders.

In clinical practice, polysomnography (PSG) is often used as the basis for gathering sleep staging data [4], encompassing electroencephalography (EEG), electrooculography (EOG), and electromyography (EMG). Human experts divide PSG recordings into 30-s epochs and perform manual scoring following either the Rechtschaffen and Kales (R&K) [5] or the American Academy of Sleep Medicine (AASM) guidelines [6]. However, manual interpretation of a single night's sleep record typically requires several hours and is subject to inter-scorer variability [7]. These limitations have motivated researchers to develop automated sleep staging methods to improve both efficiency and consistency [8]. Most early attempts to automate sleep-stage classification relied on machine learning tools, which required the design of feature extraction rules based on prior knowledge, combined with classifiers such as Naive Bayes [9], K-nearest neighbor(KNN) [10], support vector machine(SVM) [11], and random forest(RF) [12]. Although these traditional methods have achieved some success, their generalizability is limited by fixed feature extraction techniques.

With the rapid development of artificial intelligence technology, deep learning has enabled automatic feature extraction by utilizing large quantities of neurons. Accordingly, deep learning is widely used in sleep stage classification [13–18], with methods using convolutional neural networks(CNNs) [19], recurrent neural networks (RNNs) [20], and hybrid networks that combine multiple architectures [13]. By automatically extracting feature representations from signals, deep learning techniques eliminate the need for feature engineering. Attempts

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have also been made to use transformation methods such as short-time Fourier transform (STFT) [21] and continuous wavelet transform (CWT) [22] to obtain the frequency-domain modal representations of original signals that is, time–frequency images for use as input data. Based on time–frequency images, SeqSleepNet [23] constructs a hierarchical RNN to separately handle short- and long-term features. SleepTransformer [21] uses a Transformer to provide interpretability for input time–frequency images. Although these methods have improved the performance of sleep staging, each of them typically utilizes only one type of feature in the time-domain signal or frequency-domain image. This single-modality approach limits models from achieving optimal performance.

Multimodal methods can utilize information from different sources to achieve better results than those using single data sources. Owing to the diversity of data sources, these models have been widely explored in various fields, particularly in the biomedical field [24,25]. In sleep stage classification, initially, for example, EpochNet [26] uses the local pattern transformation (LPT) method to convert PSG signals into new signals containing local patterns and fuses features from multiple channels. Additionally, studies have increasingly adopted the original one-dimensional signal and transformed two-dimensional images as multimodal inputs [27–30]. For example, XSleepNet [27] learns joint representations from raw signals and time–frequency images while preserving different modal feature representations. SENDAL [28] recognizes the heterogeneity of multimodal physiological signals, employing domain adversary learning technology to construct a multimodal signal representation. MixSleepNet [29] captures the spatial, temporal, and frequency features of a multichannel PSG using 3D and graph convolutions. Although the aforementioned attempts have achieved remarkable results in sleep stage classification, they generally overlooked the interaction between different modalities and did not fully leverage the rich information of these modalities.

To utilize the complementary information of different modalities of PSG signals and thereby achieve high-performance sleep staging, we propose CrossFusionSleepNet as a deep-learning model that uses a parallel network to process multimodal inputs, facilitating the interaction of feature information between different modes. CrossFusionSleepNet can adaptively extract features between various channels and integrate multimodal feature information to provide accurate staging. The main contributions of this study are summarized as follows :

1. Multimodal sleep staging method is proposed based on PSG signals, mitigating the problem of insufficient single-modal information by using time–frequency transformations, enabling the model to simultaneously utilize the multimodal features of one-dimensional signals and two-dimensional time–frequency images.

2. A feature alignment module is designed to address the inconsistencies between the time- and frequency-domain inputs. A parallel encoder architecture comprising an attention mechanism is adopted. Moreover, input alignment, symmetric branching, parameter sharing, and other designs are implemented to efficiently extract multichannel features within the modalities.

3. A cross-attention mechanism is employed to enable intermodal feature interactions. Bidirectional content fusion can facilitate the complementary fusion of different modal information and gradually ensure excellent sleep staging performance.

2. Materials

2.1. Datasets

To thoroughly evaluate the performance of the proposed model, following previous studies [13,31], we conducted experiments on three open data sets: SleepEDF-20 [32,33], SleepEDF-78 [32,33], and Sleep Heart Health Study (SHHS) [34,35]. Dataset information is summarized in Table 1.

SleepEDF-20: SleepEDF-20 is a common baseline dataset used to determine sleep stage, encompassing data from two studies [32,33]: the Sleep Cassette (SC) study involving 20 participants between the ages of 25 and 34 yr, designed to examine the relationship between age and sleep in healthy subjects, and the Sleep Telemetry (ST) study, which examined the effects of temazepam on sleep. Following previous studies [13], we used only the SC subset to avoid other factors. The dataset contains two consecutive overnight PSG records for each participant; however, data for one night of Participant 13 was lost due to an instrument failure. Sleep experts manually labeled each 30-s PSG according to the R&K criteria [5] into eight categories {W,N1,N2,N3,N4,REM, MOVEMENT, UNKNOWN}. According to the American Academy of Sleep Medicine (AASM) [6], the R&K sleep stage division rule was modified and supplemented; the N3 and N4 stages were unified into N3 stages, and MOVEMENT and UNKNOWN were excluded [27], and 30 min wake stages both before and after sleep are used. Each epoch recorded by PSG was divided into five stages. We used the Fpz-Cz EEG, Pz-Oz EEG, and ROC-EOG(horizontal) channels with a sampling rate of 100 Hz for sleep staging.

SleepEDF-78: SleepEDF-78 is an extended version of Sleep-EDF, encompassing data from 78 healthy white participants aged 25–101 yr [32,33]. As in SleepEDF-20, the PSG was recorded for each participant for two nights. Each of Participants 13, 36, and 52 had one record lost due to an instrument error. The database was expanded to include 197 PSG with hypnosis charts, criteria for manual scoring, and labels for the 30-s PSG period, which are the same as in SleepEDF-20. Sleep staging was performed using Fpz-Cz EEG, Pz-Oz EEG, and ROC-EOG (horizontal) channels, wake stages, including the 30 min periods before and after sleep, were processed according to the Sleep-EDF-20 protocol.

SHHS: The SHHS was a multicenter cohort study designed to determine the effects of sleep-breathing disorders on cardiovascular and other health outcomes, investigating whether these disorders are associated with an increased risk of cardiovascular disease [34,35]. The dataset contains a two-part PSG record, with SHHS1 recording 5793 participants aged 40 yr and older and SHHS2 recording 2651 participants who underwent a second PSG. Keeping the same settings as in previous studies [36] the experiment incorporated the complete set of available records from the dataset. Using the same setup as the other datasets, N3 was merged from the previous N3 and N4 phases, while the MOVEMENT and UNKNOWN phases were excluded. Sleep stages were assessed using the C4-A1 EEG, C3-A2 EEG, LOC EOG, and ROC EOG channels.

2.2. Data preprocessing

As described in Section 1, CrossFusionSleepNet utilizes both time- and frequency-domain features as inputs. Each raw signal sampled at 100 Hz is segmented into a five-stage 30-s epoch; the time-domain input is $3000 \times C$, where C denotes the utilized channel. The time-domain signal is normalized to a zero mean and unit variance. To obtain the input in the frequency domain, the 30-s PSG epoch $X^S = \left\{XS_{eeg1}, XS_{eeg2}, XS_{eog}, \dots\right\}$, $X_i^S \in \mathbb{R}^{3000\times C}$ is divided into two-s windows with 50% overlap multiplied by Hamming windows and transformed into the frequency domain using a 256-point STFT. The amplitude spectrum is then transformed logarithmically, and the obtained time–frequency image input shape is $X^T \in \mathbb{R}^{T\times F\times C}$ with $F = 128$ frequency bins and $T = 29$ time points. The extracted time–frequency image is likewise normalized to a zero mean and unit variance.

3. Methods

The efficient extraction and utilization of multimodal complementary information and one-to-one architecture lie at the core of the proposed CrossFusionSleepNet design. The overall architecture is illustrated in Fig. 1. The main body of the model is composed of two parallel transformer encoders [35]. One path extracts features from the

Table 1
Details of datasets used for experiments.

Datasets	Subjects	Sampling rate	Sleep stage					Total samples
			W	N1	N2	N3	REM	
SleepEDF-20	20	100 Hz	8285 (19.60%)	2804 (6.60%)	17,799 (42.10%)	5703 (13.50%)	7717 (18.20%)	42,308
SleepEDF-78	78	100 Hz	66,822 (34.00%)	21,522 (11.00%)	69,132 (35.20%)	13,039 (6.60%)	25,835 (13.20%)	196,350
SHHS	5793	125 Hz	1,691,288 (28.85%)	217,583 (3.71%)	2,397,460 (40.89%)	739,403 (12.61%)	817,473 (13.94%)	5,863,207

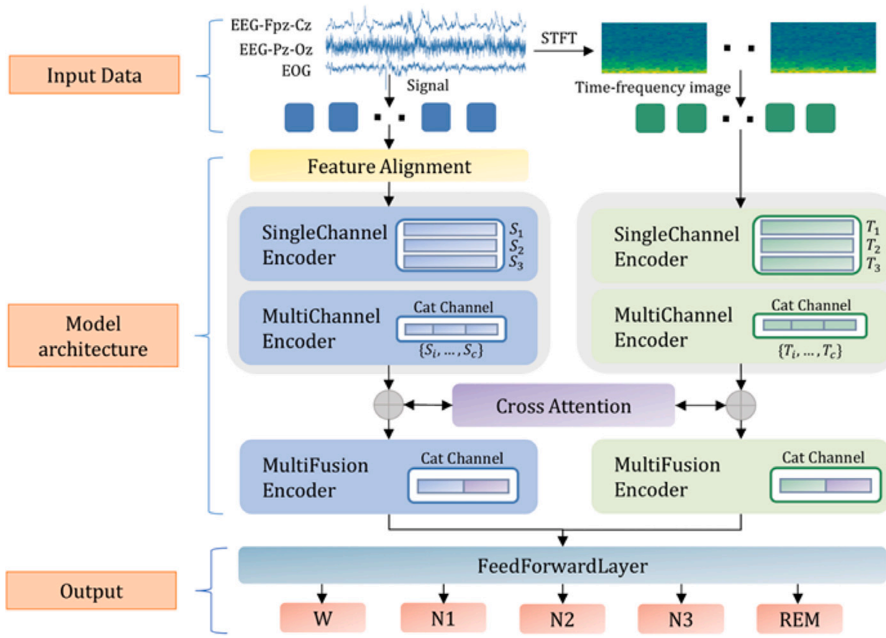

Fig. 1. CrossFusionSleepNet general schematic. Multimodal input is described, including signal input and time–frequency graph input, Model architecture details, and final output.

time-domain input, including a feature alignment module that aligns the timing- and frequency-domain features, and the other extracts features from the frequency-domain input. Following single-channel feature extraction and multichannel feature fusion, the cross-attention mechanism is used to fuse the multimodal information from different branches.

3.1. Features alignment

As described in Section 2, differences exist in the initial dimensions of the time- and frequency-domain features. To facilitate network parallelization, we designed a convolutional feature-alignment module to transform the time-domain features into the same dimensions as the frequency-domain features, as shown in Fig. 2. Given the PSG signal X^S, features are initially extracted using convolution kernels of different sizes. Each convolution layer includes a batch normalization layer, which is activated using a Gaussian error linear unit function (GELU). A max-pooling layer is used in the middle of the module for downsampling, and dropout is used to prevent overfitting.

The squeeze and excitation (SE) residual network [37] is then used to compress the channel information. The SE block selects discriminating features by adaptively modeling the channel dependency, ensuring that the time-domain features align with the frequency-domain features. Following the Conv1D residual SE block with two layers of convolution kernels of size 1, the 128 channels are compressed to 29 channels. Following one AdaptiveAvgPool1D layer and two fully connected layers, the time-domain features share a dimensionality with the frequency-domain features. The specific parameters and feature shapes are shown in Fig. 2.

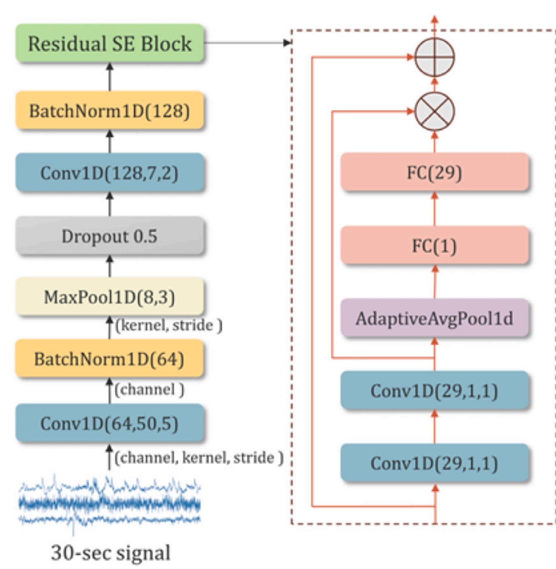

Fig. 2. Feature alignment module. The parameter meanings of the different layers are shown in the figure and GELU activation is applied after each convolutional layer.

3.2. Single-channel feature extraction and multi-channel feature fusion

The proposed network extracts the features of each channel to obtain the joint features of multiple channels. A Transformer [38] is a sequence-to-sequence model, generally composed of encoders and decoders. Encoders convert sequences into abstract representations, whereas decoders are often used for generation or prediction tasks. As the goal is to achieve sleep stage classification, only the encoder weighs the feature vectors, which comprise two core modules: multi-head attention and a feedforward network. Multi-head self-attention is correlated with elements at different locations in the input sequence. It calculates the weighted sum by first applying H learnable linear projections to the input; mapping them to the query, key, and value; calculating the scaling dot-product attention; and then concatenating the results. This process can be expressed as follows:

$$Q_i = XW_i^Q, K_i = XW_i^K, V_i = XW_i^V, 1 \le i \le H \tag{1}$$

$$H_i = Attention\left(Q_i, K_i, V_i\right) = softmax\left(\frac{Q_i K_i^T}{\sqrt{d}}\right) V_i \tag{2}$$

$$\tilde{X} = Concat\left(H_1, \ldots, H_H\right) W^X \tag{3}$$

Here, $X \in \mathbb{R}^{l \times d}$ represents an input of length l and dimension d; $Q_i, K_i, V_i \in \mathbb{R}^{l \times \frac{d}{H}}$ represents the query, key, and value mappings, respectively; $H_i \in \mathbb{R}^{l \times \frac{d}{H}}$ refers to the ith attention head; $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{d \times \frac{d}{H}}$, and $W^X \in \mathbb{R}^{d \times \mathrm{d}}$ comprise a weighting matrix for learning; and $\tilde{X} \in \mathbb{R}^{l \times \mathrm{d}}$ is the note output.

A feedforward network (FFN) provides nonlinear feature representations using two fully connected layers with ReLU activation:

$$FFN(X) = max\left(0, XW_1 + b_1\right) W_2 + b_2 \tag{4}$$

Here, W_1, W_2, b_1 and b_2 represent the learnable weights and biases, respectively. We used a transformer encoder to encode both the time and frequency domains features, and the PSG features of each channel passed through N_S encoders:

$$Z_i = Encoder\left(Z_{i-1}\right), 1 \le i - 1 < i \le N_S \tag{5}$$

where $Z_0 = \tilde{X}$, Z_i represents the feature map of each channel following feature extraction. In each time- and frequency-domain encoder, all channels share the parameters of the same encoder layer.

Temporal information (location) is vital for classifying sleep stages [21]. As attention does not contain sequential details, we introduced location information via absolute position coding. Each position index is encoded as the vector $p_t = PE(t) \in \mathbb{R}^{T \times F}$, with trigonometric functions used to encode the positional information.

$$PE_{(pos,2i)} = sin\left(pos/1000^{2i/F}\right), \tag{6}$$

$$PE_{(pos,2i+1)} = cos\left(pos/1000^{2i/F}\right), \tag{7}$$

where pos denotes the position, $0 \le pos \le T$, $2i$ or $2i{+}1$ is the dimension of the input, and $0 \le 2i \le 2i + 1 \le F$.

Combining the features of different channels is convenient for the subsequent fusion of different modal features. We combined multiple single-channel features along the length direction to obtain combined multichannel feature information. Overfitting is reduced by dropout and normalized using LayerNorm. In addition, the position of the input before embedding is connected with the encoder output. Residual connection preserves the individual features of each channel to a certain extent while alleviating the gradient vanishing problem during training. These operations can be formulated as

$$\tilde{M} = M + PE, \tag{8}$$

$$\tilde{Z}_i = Encoder\left(\tilde{M}_{i-1}\right), 1 \le i - 1 < i \le N_M \tag{9}$$

$$\tilde{O} = LayerNorm\left(\tilde{M} + \tilde{Z}_{N_M}\right), \tag{10}$$

where PE denotes positional coding and $\tilde{O} \in \mathbb{R}^{L \times CD}$ is a multichannel feature within the different modes extracted by the block, with $L = 29$, $C = 3$, $D = 128$.

3.3. Multimodal information interaction and classification

We introduced a cross-attention mechanism to address the challenges presented by feature distinctions in the time and frequency domains, thereby achieving deep multimodal feature fusion. Fig. 3 illustrates the structure of the cross-attention block, which is similar to multi-head attention except that the input time domain $Z_S \in \mathbb{R}^{L \times CD}$ and frequency domain $Z_T \in \mathbb{R}^{L \times CD}$ features are the different Q, K, and V values of the cross-attention, representing an association between two different modes. By calculating the degree of correlation between the time- and frequency-domain feature, information interaction and reweighting are realized for different modality features.

$$Q_i = Z_S W_i^Q, K_i = Z_T W_i^K, V_i = Z_T W_i^V, 1 \le i \le H, \tag{11}$$

$$\begin{aligned} H_i &= CrossAttention\left(Q_i, K_i, V_i\right) \\ &= softmax\left(\frac{Q_i K_i^T}{\sqrt{d}}\right) V_i, \end{aligned} \tag{12}$$

$$O_S = Concat\left(Z_S, H_i\right), O_T = Concat\left(Z_T, H_i\right), \tag{13}$$

The result is then mapped from a higher dimension to the same dimension as that of the input $Z \in \mathbb{R}^{L \times CD}$ through another fully connected layer. Finally, the output results should be associated with Z_S and Z_T in the channel direction to obtain O_S and O_T, respectively, which provide the differences between the different modes while preserving the original modal information. To fully integrate the different modal features, the same structure as the multichannel feature fusion module is employed, and the transformer encoder N_M is reused:

$$O_i = Encoder\left(O_{i-1}\right), 1 \le i - 1 < i \le N_M \tag{14}$$

Here, O_0 is O_S or O_T, and $O_{N_M} \in \mathbb{R}^{L \times CD}$ is the feature map of the mode after feature extraction from the encoder. Finally, extracted feature information passes through two fully connected layers to classify sleep stages. The first fully connected layer contains 1024 neurons with ReLU activation, with a dropout rate set to 0.1. The second fully connected layer contains five neurons and uses the softmax function to obtain the class probability.

3.4. Experimental parameters

The experiment was conducted on the Ubuntu 18.04 operating system with Python 3.8 programming language and the PyTorch 2.12 deep learning framework. The training process was based on the GeForce RTX 4090 GPU and 11.8 CUDA toolkit. AdamW [39] optimizer was used with the cross entropy loss function, learning rate of 5×10^{-6}, and weight decay of 0.01. An early stop strategy was introduced during training, with training stopping if the accuracy did not increase after more than 20 epochs. For consistency with previous studies [13,17,30,31], SleepEDF-20 and SleepEDF-78 adopt 20-fold cross-validation (leave-one-subject-out cross validation) and 10-fold cross-validation respectively. For the SHHS dataset, we randomly split the subjects in the dataset into 70% for training and 30% for testing, with 100 subjects from the training set separated as a validation set. The model structure and parameters in the different branches were consistent, except for the feature alignment module.

3.5. Performance metrics

Four metrics were used to evaluate model performance in detail: accuracy (ACC), macro average F1 score (MF1), and Cohen kappa (κ), with F1 used for each category. MF1 represents the arithmetic average of F1 scores for the five categories. For class i, given true positives (TP_i), false positives (FP_i), true negatives (TN_i), and false negatives (FN_i), the ACC and MF1 scores are defined as follows:

$$ACC = \frac{\sum_{i=1}^{K} T}{M}, \tag{15}$$

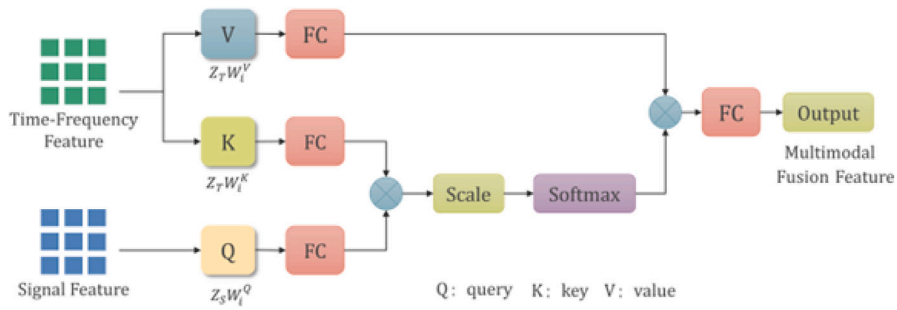

Fig. 3. Cross attention. The time–frequency feature is the frequency domain branch feature, and the signal feature is the time domain branch feature. Finally, the fused multimodal features with the same input size are obtained.

$$Precision_i = \frac{TP_i}{TP_i + FP_i}, \tag{16}$$

$$Recall_i = \frac{TP_i}{TP_i + FN_i}, \tag{17}$$

$$MF1 = \frac{1}{K}\sum_{i=1}^{K}\frac{2 \times Precision_i \times \mathrm{Recall}_i}{Precision_i + \mathrm{Recall}_i}, \tag{18}$$

M indicates the total number of samples, and K indicates the number of classes or sleep stages.

3.6. Compare with previous studies

This section presents a comprehensive performance comparison of the proposed method along with recent advanced methods. As shown in Table 2, CrossFusionSleepNet demonstrated better performance than the listed methods in multiple sleep stages on both SleepEDF-20 and SleepEDF-78 datasets, particularly in the N1 stage of the SleepEDF-20 dataset, with a 8.1% improvement in F1 score over the previous best overall performance of MultiChannelSleepNet [31]; however, SleepBoost [40] achieved the optimal F1 score on N3. On the large-scale SHHS dataset, CrossFusionSleepNet achieved the second-best performance, trailing behind CoRe-Sleep [30]. This performance gap may be attributed to the fact that CoRe-Sleep was specifically fine-tuned on the SHHS dataset, whereas CrossFusionSleepNet was optimized on a different dataset (Sleep-78) and applied to SHHS without additional adaptation. Despite the absence of SHHS-specific tuning, CrossFusionSleepNet still demonstrated competitive performance, highlighting its generalizability across datasets. Supplementary Material Fig. S1 shows the confusion matrix of CrossFusionSleepNet on the SleepEDF-20, SleepEDF-78, and SHHS datasets.

Fig. 4 shows the sleep map of subject SC4061E0 from the SleepEDF-20 dataset. The results predicted by the model are similar to the ground truth. For this subject, the accuracy of the CrossFusionSleepNet classification was 92.5%, and the corresponding F1 score was 0.88. Most misclassified periods were in the N1 stage, which is consistent with the results encapsulated in Table 2.

3.7. Multimodal result analysis

To verify the effectiveness of our multimodal approach and variability in results between the different modalities, we conducted ablation experiments and the results are shown in Fig. 5. Compared with the complete CrossFusionSleepNet model, a configuration that used only single-modal features from the three channels of the signal (Only Signal) exhibited an accuracy decrease of 4.65%, along with an average MF1 decrease of 6.46%. The model configuration that used only the unimodal features from the three channels of the frequency-domain image (Only TF-Image) exhibited a 1.54% decrease in accuracy and a 0.96% decrease in MF1. Another CrossFusionSleepNet configuration that did not use the cross-attention mechanism (No Cross Attention) exhibited a 0.49% decrease in accuracy and 0.61% decrease in MF1.

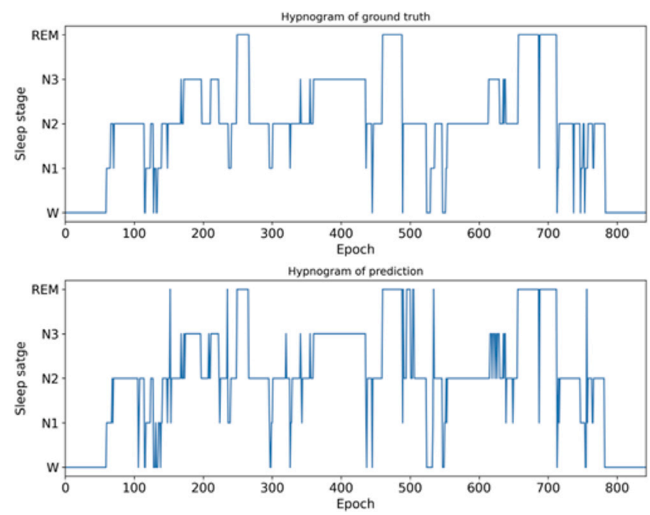

Fig. 4. Expert manual classification of sleep maps (top row) and model automatic classification of sleep maps (bottom row) from sleepEDF-20 subject SC4061E0.

Specifically, we conducted experiments on single- and multimodal inputs to investigate the impact of different channels on model performance under multimodality. The results summarized in Table 3 indicate that model performance is quasi-linear related to the number of PSG signal channels for the same input modality (single-modal signal, single-modal TF image, and multimodality). Furthermore, model performance was similar for the same number of channel inputs. For the same input channels, multimodal performance was 4.7% and 1.6% higher than that single-modal performance (Signal input and TF Image input) in terms of accuracy, and 6.4% and 0.9% higher in terms of MF1. In addition, the frequency-domain modality outperformed the time-domain modality, and the multimodal dual-channel configuration outperformed the single-modal three-channel configuration. Overall, the multichannel and multimodal configurations achieved the best performance.

3.8. Parameter analysis

To determine the impact of the network module parameters on performance, we set different values of (number of transformer encoders for single-channel feature extraction), (number of transformer encoders for multichannel feature fusion), and H (number of settings) to determine the performance of CrossFusionSleepNet on the SleepEDF-78 dataset. The model was evaluated using three indicators: accuracy, Cohen's kappa (κ), and MF1. In each experiment, any parameters not being tested were fixed. As indicated in Tables S1 and S2 of the Supplementary Materials, the combination of N_S =6 and N_M =6 achieved optimal results in terms of parameter quantity and model performance.

Table 2
Overall experimental results of CrossFusionSleepNet with previous studies.

Dataset	Method	Overall metrics			Per-class F1-score				
		Acc	MF1	κ	W	N1	N2	N3	REM
SleepEDF-20	CrossFusionSleepNet*	**89.0**	**84.0**	**0.85**	**94.1**	**57.2**	**91.3**	89.9	**87.8**
	MultiChannelSleepNet[b] [31]	87.2	81.2	0.82	92.8	49.1	90.0	89.3	84.8
	XSleepNet[a] [27]	86.4	80.9	0.81	91.5	51.1	87.5	86.9	87.4
	SleepBoost[a] [40]	86.3	80.9	0.81	89.1	50.3	90.3	**91.0**	79.5
	SeqSleepNet[b] [23]	86.0	79.7	0.81	91.9	47.8	87.2	85.7	86.2
	EpochNet [41]	84.8	78.2	0.79	89.7	44.4	88.3	84.9	83.4
	AttnSleep [13]	84.4	78.1	0.79	89.7	42.6	88.8	90.2	79.0
SleepEDF-78	CrossFusionSleepNet*	**87.5**	**83.0**	**0.83**	**95.4**	**59.4**	**88.6**	**85.5**	86.3
	MultiChannelSleepNet[b] [31]	85.0	79.6	0.79	94.0	53.0	86.9	81.8	82.6
	SCMT-15 [17]	84.3	79.4	0.79	93.5	54.3	85.5	77.0	**86.7**
	XSleepNet[a] [27]	84.0	78.7	0.78	92.6	50.3	85.5	79.2	85.7
	SeqSleepNet[b] [23]	83.8	78.2	0.78	92.8	48.9	85.4	78.6	85.1
	AttnSleep [13]	81.3	75.3	0.74	92.0	42.0	85.0	82.1	74.1
	SleepTransformer[b] [21]	81.4	74.3	0.74	91.7	40.4	84.3	77.9	77.2
	LGBT [26]	81.0	77.7	0.73	87.9	54.9	83.6	76.1	85.9
	SleepEEGNet [42]	80.0	73.6	0.73	91.7	44.0	82.5	73.5	76.1
SHHS	CrossFusionSleepNet*	88.9	81.8	0.84	**93.7**	51.2	**89.5**	84.6	**89.9**
	CoRe-Sleep[a] [30]	**89.5**	**82.3**	**0.85**	–	–	–	–	–
	L-SeqSleepNet[b] [43]	88.4	81.4	0.84	93.1	51.1	89.0	**84.9**	89.8
	SCMT-15 [17]	87.7	81.4	0.83	92.8	**52.5**	88.3	83.7	89.8
	FullSleepNet [44]	87.5	80.3	0.83	92.8	48.5	88.0	82.4	89.8

Those not marked use signal inputs.
[a] Indicates the method uses multimodal PSG inputs.
[b] Indicates the method uses time–frequency inputs.

Table 3
Performance of CrossFusionSleepNet on SleepEDF-78 dataset for combinations of different channels.

Modal	Input channel	Overall metrics			Per-class F1-score				
		Acc	MF1	κ	W	N1	N2	N3	REM
Signal	Fpz-Cz	80.5	72.8	0.73	90.1	38.0	84.3	80.5	71.2
	Fpz-Cz + Pz-Oz	81.2	74.7	0.74	92.0	42.2	84.8	82.2	72.4
	Fpz-Cz + EOG	81.7	75.2	0.75	91.5	43.2	84.6	81.4	75.2
	Fpz-Cz + Pz-Oz + EOG	**82.8**	**76.6**	**0.76**	**93.3**	**44.5**	**85.6**	**83.4**	**76.0**
TF Image	Fpz-Cz	81.7	74.8	0.75	92.1	43.1	84.7	79.4	74.7
	Fpz-Cz + Pz-Oz	85.1	80.0	0.81	95.3	53.8	85.0	83.6	82.5
	Fpz-Cz + EOG	85.3	80.8	0.81	94.8	55.5	85.6	83.1	84.8
	Fpz-Cz + Pz-Oz + EOG	**85.9**	**82.1**	**0.82**	**95.4**	**58.8**	**87.7**	**83.7**	**85.4**
Signal and TF Image	Fpz-Cz	83.8	76.4	0.77	93.6	45.0	86.4	81.5	75.6
	Fpz-Cz + Pz-Oz	86.6	81.5	0.81	**95.4**	56.0	87.9	84.5	83.8
	Fpz-Cz + EOG	86.7	81.8	0.82	94.8	56.4	88.1	84.3	85.4
	Fpz-Cz + Pz-Oz + EOG	**87.5**	**83.0**	**0.83**	**95.4**	**59.4**	**88.6**	**85.5**	**86.3**

Table 4
Performance of CrossFusionSleepNet input over three epochs on SleepEDF-78 dataset.

Method	Number of epoch	Overall metrics			Per-class F1-score				
		Acc	MF1	κ	W	N1	N2	N3	REM
CrossFusionSleepNet	3	**89.2**	**85.1**	**0.85**	**95.9**	**63.7**	**89.9**	85.3	**90.7**
CrossFusionSleepNet	1	87.5	83.0	0.83	95.4	59.4	88.6	**85.5**	86.3
AttnSleep	3	82.9	78.1	0.77	92.6	47.4	85.5	83.7	81.5
SeqSleepNet	3	82.6	76.3	0.76	91.8	46.0	85.0	77.5	81.0

We therefore selected 6 as the number of encoders for feature extraction and fusion.

To evaluate the long-term classification performance of the model, we captured correlations between consecutive epochs. Following previous research settings [13,23], we changed the CrossFusionSleepNet input to three consecutive epochs and predicted the mid-epoch labels. The results are summarized in Table 4. On the SleepEDF-78 dataset, the CrossFusionSleepNet multi-epoch input improved accuracy by 1.7% compared to the single-epoch input, with a corresponding MF1 improvement of 2.1%.

4. Discussion

This study was conducted to harness multimodal information from PSG to develop a high-accuracy sleep stage classification model. The proposed CrossFusionSleepNet integrates complementary information from both the time and frequency domains, thereby performing effectively on three publicly available datasets: SleepEDF-20, SleepEDF-78, and SHHS.

Although the automation of sleep classification [45] has become more efficient owing to machine learning, it continues to rely on prior knowledge for feature extraction. As in previous studies [13,23,31], we employed end-to-end deep learning methods for automatic feature extraction and classification. A detailed performance comparison with recent advanced deep learning methods is presented in Table 2.

AttnSleep [13] integrates a multiresolution convolutional neural network (MRCNN), an adaptive feature recalibration (AFR) module, and a temporal context encoder (TCE). By contrast, the proposed CrossFusionSleepNet employs a parallel transformer encoder branch with a streamlined architecture. On the two public datasets, SleepEDF-20 and

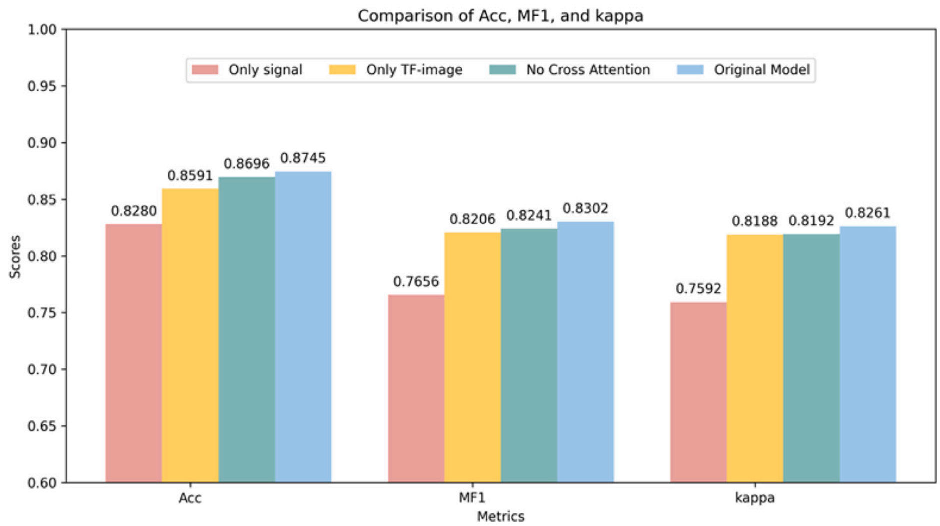

Fig. 5. Results of ablation experiments. Only signal (red) means only using the features provided by the combination of the three signal channels. Only TF-image (yellow) means only using features provided by the combination of the three channels of the frequency domain image. No Cross Attention (green) means using the features provided by the combination of the three channels of the signal and the frequency domain image, but without using Cross Attention. CrossFusionSleepNet (blue) means using the complete model.

SleepEDF-78. CrossFusionSleepNet achieved accuracy improvements of 4.6%, 5.9%, along with corresponding MF1 improvements of 5.9%, 7.7%. MultiChannelSleepNet [31] utilizes transformed time–frequency images as inputs and employs a transformer encoder for sleep staging. Contrastingly, CrossFusionSleepNet uses multimodal and multichannel inputs along with network parallelization to process diverse modal inputs, effectively leveraging the feature representations of each modality. As listed in Table 2, on the public datasets SleepEDF-20 and SleepEDF-78, CrossFusionSleepNet outperformed MultiChannelSleepNet [31] by 1.8%, 2.5%, respectively, in terms of accuracy, alongside improvements of 2.8%, 3.4%, respectively, in MF1. As shown in Table 2, CoRe-sleep, which adopts a multimodal approach, achieved the best performance on the SHHS dataset. Similarly, our proposed model, CrossFusionSleepNet, attained state-of-the-art results on the SleepEDF-20 and SleepEDF-78 datasets, also leveraging multimodal information. These consistent findings across datasets underscore the advantage of integrating multiple data modalities over unimodal approaches in sleep stage classification.

Previous studies have highlighted the potential of using multimodal inputs to construct sleep stage models. XSleepNet [27] learns joint representations from raw signals and time–frequency images, effectively capturing the underlying distribution and alleviating the problem of different views potentially generalizing or overfitting at different rates. As indicated in Table 2, compared to MultiChannelSleepNet [31] with a single-modal input, XSleepNet [27] does not fully exploit the potential of multimodal inputs, whereas the proposed CrossFusionSleepNet utilizes both multimodal and multichannel inputs. To better leverage the information between different modalities and channels, we used a parallel network to obtain the fusion of multichannel input features. Through a cross-attention mechanism, we enabled feature interaction between different modalities in the parallel network, not only alleviating the problem of interaction between time- and frequency-domain feature information, but also achieving excellent sleep staging effects. On the SleepEDF-20 and SleepEDF-78 datasets, CrossFusionSleepNet outperformed XSleepNet [27] by 2.6% and 3.5%, respectively, in terms of accuracy, alongside improvements of 3.1% and 4.3% in terms of MF1. Overall, the proposed method presents novel concepts for achieving high-accuracy sleep classification.

As demonstrated in prior studies [13,27,31], the EEG Fpz-Cz channel outperforms other channels as the base channel, as it captures a broad range of frequencies including delta activity, K-complexes, and low-frequency sleep spindles [42]. As indicated in Table 3, when the Pz-Oz or EOG channels were added, they were observed to contribute differently to different stages of sleep. For the time-domain modes, the multichannel combination improves accuracy by 2.3% and MF1 by 3.8% compared to the single-channel combination. For the frequency domain modes, the multichannel combination improves accuracy by 4.2% and MF1 by 7.3% compared to the single-channel combination. In addition, as shown in Table 2, the performance of SleepTransformer [21] based on the frequency domain input is not as good as that of multichannel input, indicating that the combination of multiple channels can capture more sleep stage features. The performance of the multimodal channel combination was particularly notable in terms of the F1 score for each sleep stage. Furthermore, the performance improvement of the multimodal combination was particularly substantial in the N1 and REM stages. Compared with the single-channel configuration, the multimodal multichannel combination increased MF1 by 14.4% in the N1 stage and 10.7% in the REM stage, as the features of these stages are complex to capture when input into a single channel. Overall, the proposed model can identify these complex stages more accurately through the complementarity of multimodal information.

To achieve efficient feature fusion and information exchange, we explored several simple fusion methods such as feature vector addition and concatenation. As indicated in Table S3 of the Supplementary Material, the use of complex fusion [36] yields suboptimal performance, indicating that complex fusion is not suitable for our model setting, and we believe that further exploration of multimodal feature fusion can further improve its performance.

Although the fusion of multimodal features improves classification performance, issues associated with feature redundancy and overfitting still require further optimization. Future research should focus on optimizing the fusion strategy of multimodal features to further enhance model generalizability. Simultaneously, different types of time–frequency domain features can be explored, and the cross-attention mechanism can be improved to more effectively integrate multimodal information.

5. Conclusion

We proposed CrossFusionSleepNet, a multimodal sleep stage classification model that leverages PSG signal and integrates complementary information from both time and frequency domains. We designed a feature alignment module and a dual-branch parallel architecture to process different modalities with the same structure. A cross-attention mechanism was designed to enable effective information exchange between modalities. CrossFusionSleepNet showed effective performance on three public datasets: SleepEDF-20, SleepEDF-78, and SHHS. The experiments results demonstrate the effectiveness of applying multimodal approaches for sleep stage classification.

CRediT authorship contribution statement

Yang Cao: Writing – review & editing, Writing – original draft, Software, Methodology. **Wei Xiang:** Supervision, Project administration, Writing – review & editing, Formal analysis. **Jie Wei:** Funding acquisition. **ShangShang Cao:** Visualization. **Xinhui Tian:** Validation. **Jiajun Zhong:** Writing – review & editing. **Xiulian Fang:** Writing – review & editing. **Bin Luo:** Resources. **He Lyu:** Writing – review & editing, Methodology, Conceptualization. **Xiangkui Li:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.bspc.2025.108538.

Data availability

Data will be made available on request.

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